

BEACON

Verifiable AI Governance for the Agent Era

v2.4 · Small edition · Apache 2.0 · No SaaS lock-in

YES-Ship AI · YES-Steady AI · YES-Recover AI

github.com/bobrapp/aigovops-beacon

aigovopsfoundation.org

[live demo](#)

Your company runs on AI. So does your risk.

Shadow models

Teams ship LLM features your governance team has never seen.

Unlogged prompts

Inputs vanish. Outputs go unverified. Receipts do not exist.

Frameworks proliferate

NIST AI RMF, EU AI Act, ISO 42001, HIPAA — 23 and counting.

Auditors arrive

Questions you cannot answer. Evidence you cannot reproduce.

**Trust is not a feature
you bolt on.**

Trust is the substrate.

Every decision an agent makes should be signed, replayable, and verifiable — by anyone, forever.

Meet Beacon.

An open-source MCP server that turns every AI decision into a signed, verifiable receipt.

SIGNED

Ed25519

Every decision sealed
cryptographically. Tamper-evident.
Replayable forever.

SCORED

23 Frameworks

NIST AI RMF, EU AI Act, ISO 42001,
HIPAA — auto-mapped from receipts.

SHIPPABLE

Apache 2.0

Three deployment shapes. One repo.
No SaaS lock-in. No vendor capture.

Six tools. One protocol.

Beacon ships as a Model Context Protocol server. Any MCP-capable agent gets governance for free.

record_decision

Sign a decision. Returns a receipt.

verify_receipt

Re-validate any receipt against its signature.

query_inventory

Search every model, decision, and dataset on file.

score_framework

Score posture against NIST, EU AI Act, ISO 42001, HIPAA.

bundle_for_auditor

Sealed evidence pack. Hand it to anyone. They verify it themselves.

replay_case

Reconstruct any decision end to end, with full provenance.

Browse the MCP server code: github.com/bobrapp/aigovops-beacon/tree/main/mcp-public

Three shapes. One repo.

Pick the deployment that fits your trust model. Switch when it changes.

1

Local desktop client

Runs on a workstation. Owned data, owned keys.

Trust: Your team only

[View code →](#)

2

Hosted MCP server

Render-deployable. Share across an enterprise.

Trust: Authenticated callers

[View code →](#)

3

Restricted public agent

Cloudflare Worker. BYO key. Read-only safe ops.

Trust: Anyone with a key

[View code →](#)

Use it in the real world.

Concrete patterns Beacon was built to handle.

Internal copilots

Sign every prompt and response. Replay any conversation when legal asks for it.

Customer-facing agents

Bundle receipts per session. Hand them to regulators on demand, without delay.

Multi-agent workflows

Log routing decisions across agents. Detect drift between runs, before users do.

High-stakes automation

Lending, healthcare, hiring — every decision auditable, verifiable, replayable.

Model inventory

Track every model in production, with provenance, owners, and a live risk class.

Audit prep

Hand the auditor a bundle they can verify themselves. No portal logins required.

See the whole loop in 70 seconds.

Twelve animated steps. Decisions in. Receipts out. Auditor bundle, sealed and signed.



animated steps

offline · self-hosted · 1.6 MB MP4

- 01 Agent emits a decision
- 02 Beacon receives the payload
- 03 Ed25519 signature applied
- 04 Receipt stored, indexed, replayable
- 05 Framework mapping auto-scored
- 06 Inventory updated, owners notified
- 07 Verification on demand by anyone
- 08 Bundle prepared for the auditor
- 09 Bundle handed off, sealed
- 10 Auditor verifies it themselves
- 11 Case replayable end to end
- 12 Trust, on the substrate

See it live: bobrapp.github.io/aigovops-beacon/walkthrough/

Hosted MCP + restricted public agent.

Two new deployment paths shipped in v2.3. Both verified live.

HOSTED MCP

mcp-public/

Deploy to Render in one click. Shareable across your enterprise. All six tools, signed receipts, auditor bundles — over HTTPS.

- Apache 2.0
- Render-ready manifests
- Auth-gated MCP endpoints
- Same six tools as the local client

RESTRICTED PUBLIC AGENT

agent/

Cloudflare Worker. BYO key. Restricted to safe, read-only operations. Run a public-facing Beacon agent at your own edge — no surprises.

- Cloudflare Worker template
- Bring-your-own API key
- Allow-listed tool surface
- Verifiable receipts, public traffic

The layer that lets a \$1M super-agent actually ship.

Big budgets buy capability.

Beacon supplies the trust they cannot ship without.

Read the brief: [SUPERAGENT.md](#)

23 frameworks built in. Yours plugs right in.

Run `score_framework`. Get a posture report per framework, with the receipts that backed every claim.

NIST AI RMF	EU AI Act	ISO/IEC 42001	ISO/IEC 23894	HIPAA	GDPR Art. 22/35
SOC 2 TSC	AIDA (Canada)	OECD AI Principles	IEEE 7000-2021	Brazil PL 2338	Korea AI Basic Act
NYC LL 144	Colorado SB24-205	California SB-53	Texas TRAIGA	UK AI WP	UN GA A/78/L.49
Singapore GenAI	Australia VAISS	China Interim	India DPDP	US AI Bill of Rights	+ your own (YAML)

Browse all 23: github.com/bobrapp/aigovops-beacon/tree/main/frameworks

Run it today.

Three commands. One repo. Apache 2.0.

```
$ git clone https://github.com/bobrapp/aigovops-beacon
$ cd aigovops-beacon
$ ./demo.sh                                # local desktop, port 8801

$ cd mcp-public && ./deploy-render.sh      # hosted MCP server
$ cd agent && wrangler deploy               # restricted public agent
```

[DEMOS.md](#)

Three deployment shapes, end to end, in under five minutes.

[SUPERAGENT.md](#)

How Beacon lets a \$1M super-agent actually ship.

[walkthrough/](#)

Twelve animated steps. Decisions in. Receipts out.

Code is on GitHub. Community is at the Foundation.

Both are open. Both are waiting.

THE CODE

aigovops-beacon

Apache 2.0. Six MCP tools. Three deployment shapes. Walkthrough included.

github.com/bobrapp/aigovops-beacon

Star • Fork • Run • Contribute

THE COMMUNITY

AI GovOps Foundation

Where practitioners share frameworks, case studies, and reference implementations — open governance, open practice.

aigovopsfoundation.org

Join • Contribute • Lead a working group

Hand the auditor a bundle. Let the tokens flow.

Beacon. Verifiable AI governance. Available now.

→ github.com/bobrapp/aigovops-beacon

→ aigovopsfoundation.org

A P P E N D I X

AIGovOps Foundation Protocol

Repository Fact-Check

Every load-bearing claim in the repo, classified and sourced.

Full document: [aigovops-fact-check.md](#)

The Protocol

Every claim in the repository gets one of four verdicts.

VERIFIED

Corroborated by an authoritative external source: an official document, regulator filing, peer-reviewed paper, or Tier-1 news outlet. Cited with URL.

SOURCED

Direct attribution to a named source (law, framework, regulator, vendor doc). The source exists and says approximately what the repo claims. Cited with URL.

ASPIRATIONAL

Describes design intent, roadmap, or self-description. Verifiable inside the repo itself (file path cited). Not externally falsifiable.

UNVERIFIABLE

Unsourced opinion, specific stat with no public source, or forward-looking projection. Flagged with what's missing.

Repository scoreboard.

Total claims audited: 46. Audit date: 2026-05-13.

18

VERIFIED

6

SOURCED

14

ASPIRATIONAL

8

UNVERIFIABLE

Full appendix: [aigovops-fact-check.md](#)

Claims 1-5 of 46

AIGovOps Foundation Protocol · audited 2026-05-13

VERDICT	CLAIM	SOURCE / EVIDENCE
VERIFIED	Licensed under the Apache 2.0 License	apache.org/licenses/LICENSE-2.0
VERIFIED	Code of Conduct: Contributor Covenant v2.1	contributor-covenant.org/version/2/1/code_of_conduct/
VERIFIED	23 governance frameworks	<code>`ls /home/user/workspace/aigovops-beacon/frameworks/*.yaml`</code> returns 24 files. <code>`index.yam...</code>
ASPIRATIONAL	No SaaS lock-in, no phone-home	<i>Stated as a design principle in</i> <code>`/home/user/workspace/aigovops-beacon/README.md`</code> and <code>`h...</code>
ASPIRATIONAL	MCP server exposes 6 tools: record_decision, verify_receipt, query_inventory, score_framework, bundle_for_auditor, replay_case	<i>Tool list enumerated in</i> <code>`/home/user/workspace/aigovops-beacon/ARCHITECTURE-BETA.md`</code> and ...

Full appendix with citations: [aigovops-fact-check.md](#)

Claims 6-10 of 46

AIGovOps Foundation Protocol · audited 2026-05-13

VERDICT	CLAIM	SOURCE / EVIDENCE
VERIFIED	ISO/IEC 42001 — International AI management system standard	iso.org/standard/42001
VERIFIED	NIST AI RMF — Federal-grade risk management, default if you're in the US	nist.gov/itl/ai-risk-management-framework
VERIFIED	EU AI Act — Binding EU law with risk tiers (prohibited, high-risk, limited-risk, minimal-risk)	artificialintelligenceact.eu/article/13/
SOURCED	NYC Local Law 144 — Automated employment decision tools	legistar.council.nyc.gov/LegislationDetail.aspx
ASPIRATIONAL	Beacon version v2.2.0" (shown in MCP initialize response)	Version string appears in the synthetic MCP session output at <code>`/home/user/workspace/aigo...</code>

Full appendix with citations: [aigovops-fact-check.md](#)

Claims 11-15 of 46

AIGovOps Foundation Protocol · audited 2026-05-13

VERDICT	CLAIM	SOURCE / EVIDENCE
ASPIRATIONAL	Demo 1: query_inventory returns 20 rows	Demo narrative in <code>`/home/user/workspace/aigovops-beacon/DEMOS.md`</code> claims "20 inventory r...
ASPIRATIONAL	Receipts signed with Ed25519; <code>`alg: Ed25519`</code> field in output	<code>`alg: "Ed25519"`</code> appears in the sample receipt JSON in <code>`/home/user/workspace/aigovops-be...</code>
ASPIRATIONAL	46 synthetic receipts covering 97 controls across 23 frameworks" (scoring/sample_report.md)	Numbers appear in <code>`/home/user/workspace/aigovops-beacon/scoring/sample_report.md`</code> , which...
UNVERIFIABLE	DNS tail finding ~3× more AI hostnames than browser extension alone	No empirical source cited. Claim appears in <code>`/home/user/workspace/aigovops-beacon/LAB.md...</code>
UNVERIFIABLE	CASB finding ~1.5× more shadow IT than DNS alone	No source cited. Appears alongside L-01 in <code>`/home/user/workspace/aigovops-beacon/LAB.md`</code> .

Full appendix with citations: [aigovops-fact-check.md](#)

Claims 16-20 of 46

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VERDICT	CLAIM	SOURCE / EVIDENCE
UNVERIFIABLE	L1 + L2 + L3 coverage layers together cover ~90% of real corporate AI usage	Percentage asserted without citation in <code>`/home/user/workspace/aigovops-beacon/LAB.md`</code> an...
UNVERIFIABLE	\$1M super-agent" positioning / "\$500K–\$2M/year contracts	Revenue projections appear in <code>`/home/user/workspace/aigovops-beacon/SUPERAGENT.md`</code> refer...
UNVERIFIABLE	KSB Agenic \$500K per-employee revenue ratio	Same sourcing gap as S-01. Internal projection only.
UNVERIFIABLE	~40% of enterprise AI usage via browser (ChatGPT/Claude/Gemini/Copilot), ~25% internal copilots, ~15% IDE, ~10% CLI, ~10% server-side	Breakdown stated in <code>`/home/user/workspace/aigovops-beacon/ARCHITECTURE-BETA.md`</code> with no ...
ASPIRATIONAL	Append-only NDJSON audit log	Append-only NDJSON format described in <code>`/home/user/workspace/aigovops-beacon/docs/ARCHIT...</code>

Full appendix with citations:
 [aigovops-fact-check.md](#)

Claims 21-25 of 46

AlGovOps Foundation Protocol · audited 2026-05-13

VERDICT	CLAIM	SOURCE / EVIDENCE
ASPIRATIONAL	Merkle root anchored to log hourly	Hourly anchoring described in <code>`/home/user/workspace/aigovops-beacon/docs/ARCHITECTURE.md...</code>
VERIFIED	RFC 8785 JSON Canonicalization Scheme (JCS) used for deterministic signing	datatracker.ietf.org/doc/html/rfc8785
VERIFIED	Zillow shut down Zestimate-driven iBuying after \$304M inventory write-down in Q3 2021	investors.zillowgroup.com/investors/news-and-events/news/news-d...
VERIFIED	Knight Capital lost \$440M in 45 minutes due to a rogue trading algorithm (August 2012)	bloomberg.com/news/articles/2012-08-02/knight-shows...
VERIFIED	Air Canada chatbot held liable — BC Civil Resolution Tribunal (2024 BCCRT 149)	cbc.ca/news/canada/british-columbia/air-cana...

Full appendix with citations: [aigovops-fact-check.md](#)

Claims 26-30 of 46

AlGovOps Foundation Protocol · audited 2026-05-13

VERDICT	CLAIM	SOURCE / EVIDENCE
VERIFIED	Samsung banned ChatGPT after employees leaked proprietary source code (May 2023)	bloomberg.com/news/articles/2023-05-02/samsung-bans...
VERIFIED	Google paused Gemini image generation of people in February 2024	reuters.com/technology/google-pause-gemini-ai-mod...
VERIFIED	iTutorGroup paid \$365,000 to settle EEOC age-discrimination suit (ADEA)	eeoc.gov/newsroom/itutorgroup-pay-365000-settl...
VERIFIED	Hong Kong deepfake video call fraud: finance employee tricked into transferring \$25M (HK\$200M)	ft.com/content/b977e8d4-664c-4ae4-8a8e-eb93b...
VERIFIED	EU AI Act Article 13 covers transparency obligations for high-risk AI systems	artificialintelligenceact.eu/article/13/

Full appendix with citations: [aigovops-fact-check.md](#)

Claims 31-35 of 46

AIGovOps Foundation Protocol
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VERDICT	CLAIM	SOURCE / EVIDENCE
SOURCED	HIPAA Security Rule applies to healthcare AI as a baseline control set	hhs.gov/hipaa/for-professionals/security/inde...
SOURCED	Colorado SB24-205 (Consumer Protections for Artificial Intelligence) enacted	leg.colorado.gov/bills/sb24-205
VERIFIED	California SB 53 (Transparency in Frontier Artificial Intelligence Act) signed September 29, 2025	gov.ca.gov/2025/09/29/governor-newsom-signs-sb-5...
ASPIRATIONAL	Control plane enforces policy-as-code with per-framework scoring rubrics stored as YAML	Architecture described in <code>`/home/user/workspace/aigovops-beacon/docs/CONTROL_PLANE.md`</code> ; ...
ASPIRATIONAL	Bundle-for-auditor tool produces a single portable ZIP containing receipts, Merkle proof, and framework mappings	Tool described in <code>`/home/user/workspace/aigovops-beacon/docs/CONTROL_PLANE.md`</code> , <code>`/home/u...</code>

Full appendix with citations:
 [aigovops-fact-check.md](#)

Claims 36-40 of 46

AlGovOps Foundation Protocol · audited 2026-05-13

VERDICT	CLAIM	SOURCE / EVIDENCE
ASPIRATIONAL	Receipt schema is append-only; records are immutable once written	<i>Immutability constraint defined in</i> <code>`/home/user/workspace/aigovops-beacon/docs/RECEIPT_SC...</code>
VERIFIED	RFC 8785 canonicalization applied before Ed25519 signing	datatracker.ietf.org/doc/html/rfc8785
ASPIRATIONAL	Studio Flow supports human-in-the-loop review gates before agent actions are committed	<i>Workflow described in</i> <code>`/home/user/workspace/aigovops-beacon/docs/STUDIO_FLOW.md`. No ext...</code>
SOURCED	OECD AI Principles referenced as a scored framework	oecd.ai/en/ai-principles
SOURCED	UN General Assembly Resolution A/78/L.49 (AI governance) referenced	documents.un.org/doc/undoc/ltd/n24/065/92/pdf/n2406592...

Full appendix with citations: [aigovops-fact-check.md](#)

Claims 41-45 of 46

AIGovOps Foundation Protocol
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VERDICT	CLAIM	SOURCE / EVIDENCE
ASPIRATIONAL	Deck describes Beacon as covering 'the full AI governance stack from shadow AI discovery to audit-ready evidence bundles'	Marketing/pitch language in `/home/user/workspace/aigovops-beacon/tools/deck/build_deck_...
ASPIRATIONAL	Deck claims 'zero vendor lock-in' and 'runs fully local'	Claim in deck slide text at `/home/user/workspace/aigovops-beacon/tools/deck/build_deck_...
UNVERIFIABLE	Launch posts reference 'the only open-source AI governance MCP server'	Superlative claim in `/home/user/workspace/aigovops-beacon/tools/launch/launch_posts.md`...
VERIFIED	Launch posts claim '23 frameworks out of the box'	Count confirmed via directory listing and `frameworks/index.yaml`. See Claim R-03.
UNVERIFIABLE	Launch posts reference 'production-ready' status	"Production-ready" asserted in `/home/user/workspace/aigovops-beacon/tools/launch/launch...

Full appendix with citations:
 [aigovops-fact-check.md](#)

Claims 46-46 of 46

AIGovOps Foundation Protocol
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VERDICT	CLAIM	SOURCE / EVIDENCE
SOURCED	GDPR Article 22 and Article 35 (DPIA) referenced as covered frameworks	gdpr-info.eu/art-22-gdpr/

Verify every claim yourself.

That is the whole point of Beacon.

The full fact-check is a single Markdown file in the repo. Click any source, follow any URL, contest any verdict.

→ [aigovops-fact-check.md](#)

→ [github.com/bobrapp/aigovops-beacon](#)